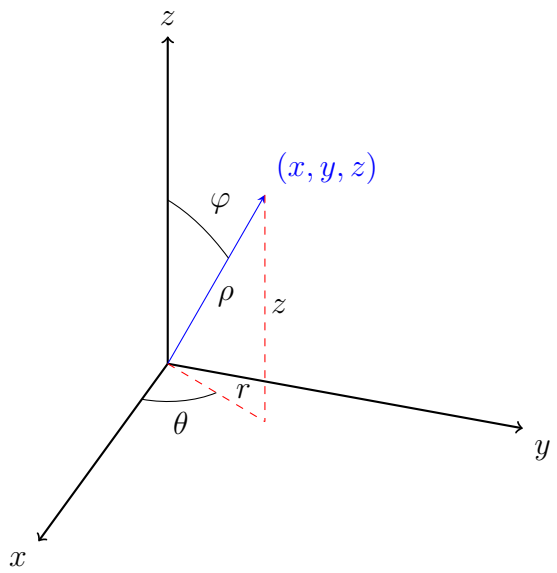
...

Triple Integrals in Spherical & Cylindrical Coordinates

Cylindrical coordinates: (r, θ, z) where (r, θ) are polar coordinates for the xy-plane.

Note: Whenever convenient, one can permute the role of z with either x, y.

Spherical coordinates: (ρ, θ, φ) where ρ is the signed distance to the origin along the ray where θ is the polar angle of the point $(x, y, 0)$ made in the xy-plane and φ is the angle between the point (x, y, z) regarded as a vector and the positive z-axis.

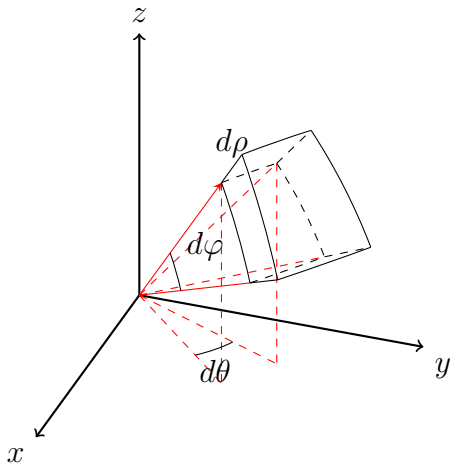


Here are some key conversions:

$\rho^2 = x^2+y^2+z^2$, $r = \rho \sin(\varphi)$, $x = r\cos(\theta) = \rho \sin(\varphi)\cos(\theta)$, $y = r\sin(\theta) = \rho\sin(\varphi)\sin(\theta)$, and $z = \rho\cos(\varphi)$.

Cylindrical coordinates are just polar coordinates plus a rectangular coordinate, so differential volume in cylindrical coordinates is $\boxed{dV = r\,dz\,dr\,d\theta}$.

For spherical coordinates, the volume ΔV of a spherical box (see picture below) can be approximated by $\Delta V \approx \Delta\rho(\rho\Delta\varphi)(\rho\sin(\varphi)\Delta\theta)$ and the error in the approximation goes towards 0 as $\Delta\rho \to 0$, $\Delta\theta \to 0$ and $\Delta\varphi \to 0$. So the differential volume of a spherical box satisfies $dV = d\rho\,(\rho d\varphi)(\rho\sin(\varphi)\,d\theta)$. Equivalently $\boxed{dV = \rho^2\sin(\varphi)\,d\rho\,d\varphi\,d\theta}$.



Let's evaluate $\iiint_E z\,dV$ where E is the part of the ball $\{(x,y,z)|x^2+y^2+z^2 \le 4\}$ above the plane $z = 1$.

First let's do it in cylindrical coordinates. When $z = 1$, $x^2 + y^2 \le 3$. Thus region can be described in cylindrical coordinates as $1 \le z \le \sqrt{4-r^2}$, $0 \le r \le \sqrt{3}$, $0 \le \theta \le 2\pi$.

$$\begin{aligned}
\iiint_E z\,dV &= \int_0^{2\pi}\int_0^{\sqrt{3}}\int_1^{\sqrt{4-r^2}} z\,r\,dz\,dr\,d\theta \\
&= 2\pi\int_0^{\sqrt{3}} \frac{1}{2}z^2 r\Big|_{z=1}^{z=\sqrt{4-r^2}}\,dr \\
&= \pi\int_0^{\sqrt{3}} (3-r^2)r\,dr \\
&= \pi\int_0^{\sqrt{3}} 3r - r^3\,dr \\
&= \pi\left(\frac{3}{2}r^2 - \frac{1}{4}r^4\Big|_0^{\sqrt{3}}\right) \\
&= \pi\left(\frac{9}{2} - \frac{9}{4}\right) = \frac{9\pi}{4}.
\end{aligned}$$

Now let's do it in spherical coordinates. The region in the ball $\{(x,y,z)|x^2+y^2+z^2 \le 4\}$ above the plane $z=1$ is equivalent to $\rho \le 2$ and $\rho\cos(\varphi) \ge 1$. So $\sec(\varphi) \le \rho \le 2$ and the bounding surfaces intersect where $\sec(\varphi)=2$, or equivalently $\cos(\varphi)=\frac{1}{2}$, implying $\varphi=\frac{\pi}{3}$. So $0 \le \varphi \le \frac{\pi}{3}$. Finally $0 \le \theta \le 2\pi$.

$$\begin{aligned}
\iiint_E z\,dV &= \int_0^{2\pi}\int_0^{\frac{\pi}{3}}\int_{\sec(\varphi)}^{2} \rho\cos(\varphi)\,\rho^2\sin(\varphi)\,d\rho\,d\varphi\,d\theta \\
&= 2\pi\int_0^{\frac{\pi}{3}}\int_{\sec(\varphi)}^{2} \rho^3\cos(\varphi)\sin(\varphi)\,d\rho\,d\varphi \\
&= 2\pi\int_0^{\frac{\pi}{3}} \frac{1}{4}\rho^4\cos(\varphi)\sin(\varphi)\Big|_{\sec(\varphi)}^{2} d\varphi \\
&= \frac{\pi}{2}\int_0^{\frac{\pi}{3}} (16-\sec^4(\varphi))\cos(\varphi)\sin(\varphi)\,d\varphi \\
&= \frac{\pi}{2}\int_0^{\frac{\pi}{3}} \left(16\cos(\varphi)\sin(\varphi) - \frac{\sin(\varphi)}{\cos^3(\varphi)}\right) d\varphi \\
&= \frac{\pi}{2}\left(-8\cos^2(\varphi) - \frac{1}{2\cos^2(\varphi)}\Big|_0^{\frac{\pi}{3}}\right) \\
&= \frac{\pi}{2}\left(-8(1/4-1) - \frac{1}{2}(4-1)\right) = \frac{9\pi}{4}.
\end{aligned}$$

Let's evaluate $\iiint_E \frac{1}{1+x^2+y^2+z^2}dV$ where E is the solid bounded by $z=\sqrt{3x^2+3y^2}$ and $z=\sqrt{25-x^2-y^2}$ where $y \ge 0$. We will do it in spherical coordinates. The cone $z=\sqrt{3}\sqrt{x^2+y^2}$ is $\rho\cos(\varphi)=\sqrt{3}\rho\sin(\varphi)$, equivalently $\varphi=\tan^{-1}(1/\sqrt{3})=\frac{\pi}{6}$ and $z=\sqrt{25-x^2-y^2}$ is the top half of the sphere $\rho=5$. Also as $y \ge 0$, $0 \le \theta \le \pi$. Hence

$$\begin{aligned}
\iiint_E \frac{1}{1+x^2+y^2+z^2}\,dV &= \int_0^{\pi}\int_0^{\frac{\pi}{6}}\int_0^{5} \frac{1}{1+\rho^2}\,\rho^2\sin(\varphi)\,d\rho\,d\varphi\,d\theta \\
&= \pi\int_0^{\frac{\pi}{6}}\int_0^{5} \left(1-\frac{1}{1+\rho^2}\right)\sin(\varphi)\,d\rho\,d\varphi \\
&= \pi\int_0^{\frac{\pi}{6}} \left(\rho-\tan^{-1}(\rho)\right)\sin(\varphi)\Big|_{\rho=0}^{\rho=5} d\varphi \\
&= \pi\int_0^{\frac{\pi}{6}} \left(5-\tan^{-1}(5)\right)\sin(\varphi)\,d\varphi \\
&= \pi\left(5-\tan^{-1}(5)\right)\left(-\cos(\varphi)|_0^{\frac{\pi}{6}}\right) \\
&= \pi\left(5-\tan^{-1}(5)\right)\left(1-\frac{\sqrt{3}}{2}\right).
\end{aligned}$$